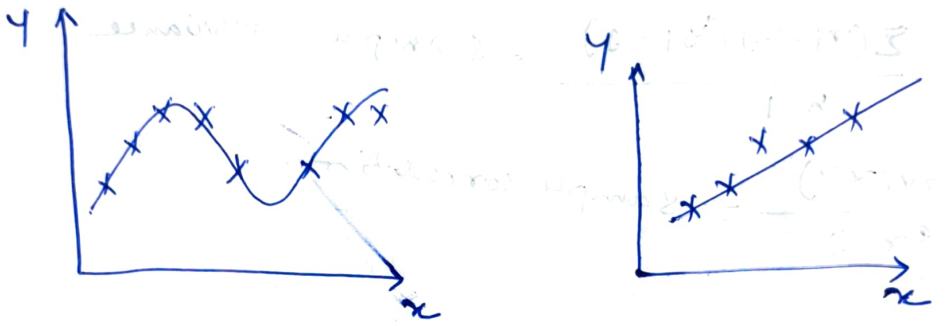
Regression

Correlation tells us if there is an association between x and y but it does not allow to predict one variable from the other.

However, regression allow us.

Start with the problem of polynomial curve fitting:



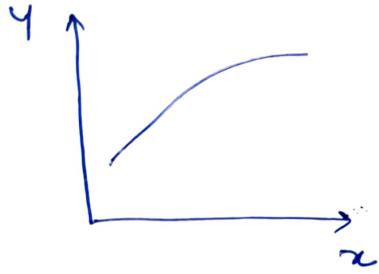
X → Independent variable

Y → dependent variable

Senario 1

If you know the equation

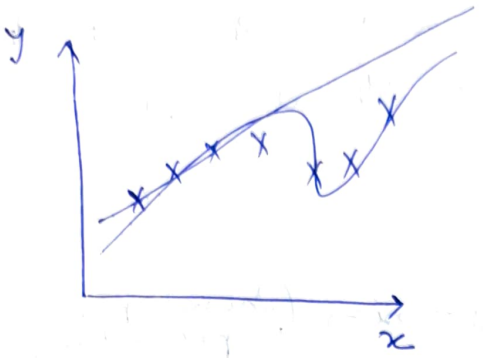
$$y = 2 + 3x + 4x^2$$



Can draw the curve

Scenario 2

- points are given.
- polynomial structure is not known
- we have to approximate the polynomial structure from some points



So many structures/ equation

Some points → training samples $\langle x_i, y_i \rangle$

x_i	y_i
0.1	5
0.6	2
0.9	1.2

Find equation (best fit one)

Example of regression problem (house price prediction)

Size of house	price
600	15
1120	30
⋮	40
⋮	⋮

Back to polynomial curve fitting problem

$$y = a_0 + a_1 x^1 + a_2 x^2 + \cdots + a_n x^n$$

or

$$y = w_0 + w_1 x^1 + w_2 x^2 + \cdots + w_n x^n$$

→ We want to find this equation from training sample $\langle x_i, y_i \rangle$

primarily unknown

n → degree of polynomial

$w_0, w_1, \ldots, w_n$ → parameters

Supervised Learning

- There is a set of training samples $\langle x_i, y_i \rangle$
- The goal is to learn the function/hypothesis $h(x_i)$ so that it is a good predictor for the corresponding value of y using training samples.

$$h : x \to y$$

$y \to$ Continuous / Regression (house price prediction)

$y \to$ discrete / Classification (spam / no-spam identification)

Aim: to design learning algorithm to approximate hypothesis structure

Linear Regression

Assume: y is a linear function of X

$$\hat{y} = h(x) = w_0 + w_1 x$$

Learn $h(x)$ from $\langle x_i, y_i \rangle$ training sample

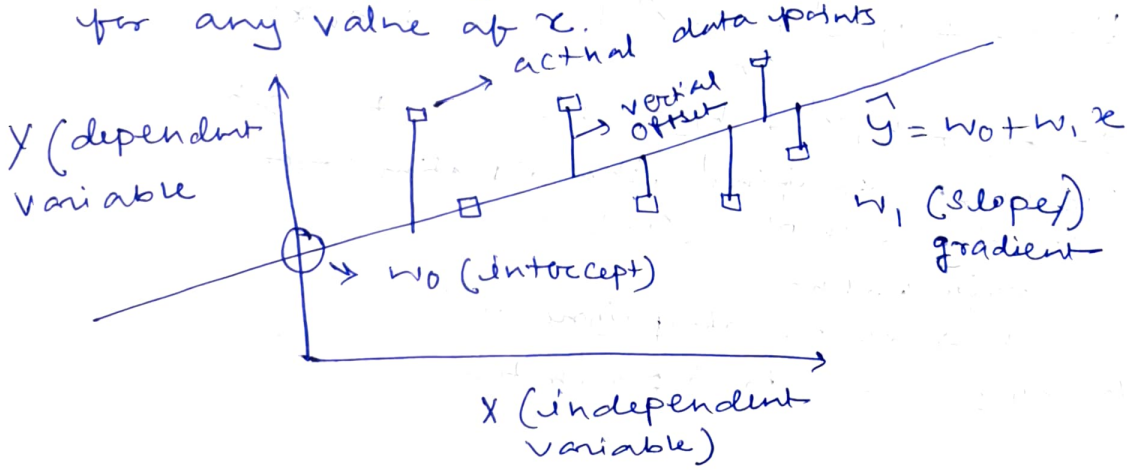
m = number of training sample

n = number of feature = 1

Aim: fit a straight line,

$$\hat{y} = h(x) = w_0 + w_1 x$$

to data that gives best prediction of y for any value of x.



Find best fit line that minimize difference/distance between data and fitted line i.e the residual (ε)

$$\varepsilon_i = y_i - \hat{y}_i \qquad i = 1, 2, \dots m.$$

y_i ↓ actual, $\hat{y}_i$ ↓ predicted

m ↓ number of training samples

- There are so many such lines or so many assumption about equation

$$\hat{y} = 2 + 3x \quad | \quad \hat{y} = 10x$$

- Find the line having minimum residual.
- For that, we have to tune w_0, w_1

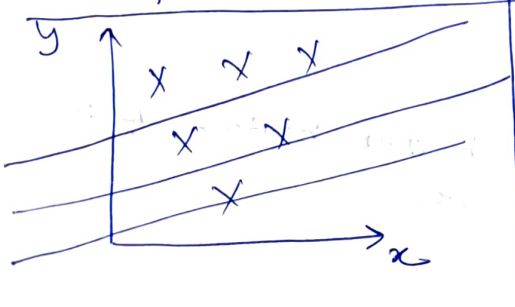
Ordinary least square regression (OLS)

- To find the best fit line that minimize the sum of the square of residuals.
- hypothesis : $\hat{y} = h(x) = w_0 + w_1 x$
- Residual : $\varepsilon = y - \hat{y}$
- The sum of square of residual = $\sum (y - \hat{y})^2$
- Find w_0, w_1 that minimize

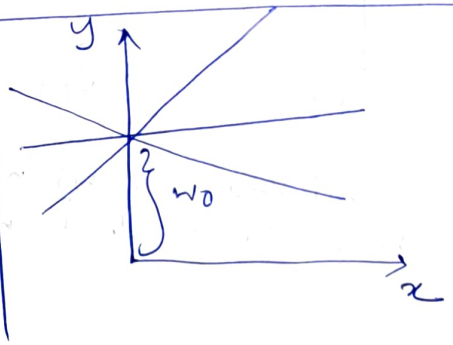
$$LSE = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

Now, we are searching for w_0, w_1

Finding w_0 (changing y intercept value)



Finding w_1 (changing slope)



Simple linear regression

One independent variable (x)
One dependent variable (y)

Model: $\hat{y} = h(x) = w_0 + w_1 x$
Residual: $(y - \hat{y})$

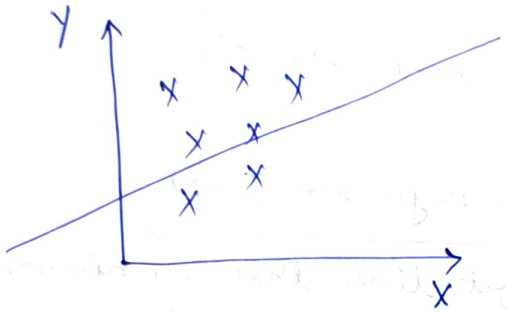
We have training samples $\langle x_i, y_i \rangle$ $i = 1, 2, \ldots m$

Minimize

$$J(w_0, w_1) = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

RSS: Residual sum of square
LSE: Least square error
$J(w)$: $w = [w_0 \; w_1]$

Find the best fitted line

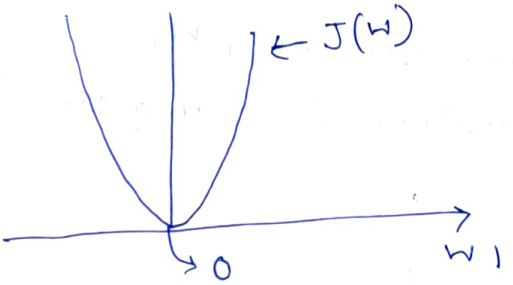


$\hat{y} = w_0 + w_1 x$
w_0: y intercept, w_1: slope

- Sometime, we use $1/2$ before LSS for convience and simplicity. It does not hamper the minimization.

$$J(w) = \frac{1}{2} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

- If we plot $J(w)$ with respect to a parameter (slope w_1) yield the classic U shapped parabola because it is a squared term.



- So, the minimum $J(w)$ is at the bottom of the curve, where the gradient is zero.

- We can find w_0, w_1 that give min $J(w_0, w_1)$ by taking partial derivative of $J(w)$ with respect to w_0, w_1 seperately.

Approch1 Ordinary linear regression (OLS)

A closed form solution in terms of linear regression, refer to an analytical method to find the exact coefficient of polynomial by minimizing the error without iterative optimization.

$$\frac{\partial}{\partial w_0} J(w) = 0 \qquad \frac{\partial}{\partial w_1} J(w) = 0$$

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x} = \frac{\sum_{i=1}^{m} y_i - \sum_{i=1}^{m} \hat{w}_1 x_i}{m}$$

$$\hat{w}_1 = \frac{cov(x,y)}{s_x^2} = r_{xy} \frac{s_y}{s_x}$$

- There is a relation between slope and correlation coefficient
- There is a relation between slope and s_x and s_y.

Multiple linear regression

One dependent variable (Y)
Multiple independent variable ($x_1, x_2, \ldots x_n$)